A3 No. and Name
Team 7

Team Leader (name & 'phone ext)

Team members (name & role)

1. Dhruv
2. Ankit
3. Jimit
4. Jay and vishal

Stakeholders (role & department)

1. AI & ML Coordinator, Conestoga College
2. Potential Client name(s)
3. Other Conestoga College stakeholder(s)
4.

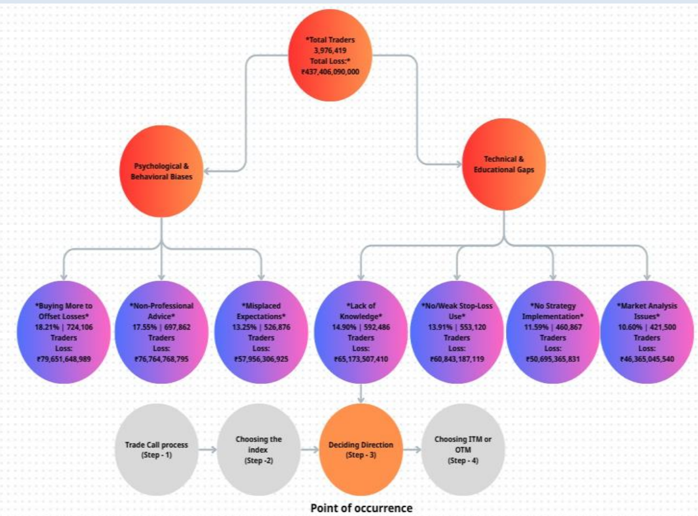
Company objective

Start date & planned duration

Your team logo and project name

1. Clarify the problem

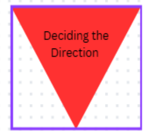
2. Breakdown the problem



3. Set the Target

1	Reduce number of options traders with lack of knowledge experiencing losses within 3 months
2	Deliver personalized trading education modules
3	Achieve a 50% increase in stop-loss compliance
4	

4. Analyse the Root Cause



1.Why do traders fail at deciding the direction of the market?
→ Because they make impulsive and emotionally driven decisions without data-backed support.
2. Why do they make impulsive and emotionally driven decisions?
→ Because they lack real-time analytical insights or structured decision-support at the time of placing a trade.
3.Why do they lack real-time decision-support?
→ Because most retail trading platforms do not offer integrated AI-powered guidance tools for directional trading.
4 Why don't platforms offer integrated AI guidance tools?
→ Because there's a legacy belief that discretionary trading is the norm, and AI interventions may be seen as limiting user autonomy.
5. Why is this legacy belief still driving platform design?
→ Because the value of AI in reducing cognitive bias and improving directional accuracy is underutilized and not embedded as a standard industry feature.

ROOT CAUSE
Because the value of AI in reducing cognitive bias and improving directional accuracy is underutilized and not embedded as a standard industry feature

5. Develop Countermeasures

			Criteria							
		Description:	ROI	Ease of use	Learning Impact	Accuracy	Scalability	User Acceptance		
		Weight (out of 100 pts):	30	10	10	30	10	10	Total(100)	Rankings
		Minimum Desired Score (/10):	7	7	8	7.5	6	7	71.5	
Options	1	AI Trade Assistant	8	6	5	9	9	5	76	3
	2	AI-Powered Onboarding Simulation	7	9	9	8	9	8	80	1
	3	AI Paper Trading Sandbox	7	8	9	8	9	8	79	2
	4	Soft AI Nudges in UI	7	9	7	7	9	6	73	4
	5	Mandatory Literacy Module	6	4	8	9	9	3	69	5

6. Implement Countermeasure

7. Monitor Results & Process

8. Standardise & Share Success

GitHub Repository and Execution Guide

All source code, datasets, Jupyter Notebooks, and execution instructions related to this project are available in the following GitHub repository:

🔗 **GitHub Link:**
https://github.com/Jimit177/AI_PREDICTOR

This repository contains:

- Jupyter Notebooks for EDA, model training, and evaluation
- Python scripts for running the backend and frontend logic
- Cleaned and labeled datasets
- Trained model files and configuration artifacts
- A Streamlit app for real-time directional predictions

📄 **Execution Instructions:**
All instructions for setting up the environment, running the EDA, performing statistical tests, and reproducing model predictions are provided in the repository's README.md file.

🧠 **Notebook LM Source:**
A link to the Notebook LM log, containing details of learning iterations and model results, is also included in the README.md.

Abstract:

The Indian equity derivatives market has witnessed an explosive rise in retail participation, with over 4.5 million individual traders engaging in Futures and Options (F\&O) during FY22—a staggering 500% increase since FY19. However, this surge has been accompanied by alarming statistics: according to SEBI's official report, *over 94% of active retail traders in the F\&O segment incurred net losses, with average losses amounting to ₹1.1 lakh per person. This report investigates the core reasons behind these losses, focusing specifically on the **knowledge-deficient sub-segment* of options traders—those who lack formal trading education, rely on social media tips, and engage in impulsive decision-making.

Using a combination of regulatory data, academic literature, and behavioral analysis, the report identifies key gaps in financial literacy, risk management, and platform design. It then evaluates a set of proposed countermeasures such as AI-powered simulations, sandbox trading environments, and behavioral nudges using a weighted OMT (One Metric That Matters) matrix. The highest-scoring interventions align directly with the learning and risk control deficits uncovered in the research.

The findings support the urgent need for *scalable, practical, and regulation-compliant solutions* that can reduce loss rates, increase trading literacy, and promote informed decision-making in India's rapidly growing retail derivatives market.

1. Introduction:

The Indian financial landscape has experienced a profound transformation in recent years, marked by the rapid growth of retail participation in the equity Futures and Options (F\&O) segment. As of FY2022, over *4.52 million unique individual traders* engaged in F\&O trading a *500% increase* from FY2019 levels largely driven by mobile trading platforms, low entry barriers, and the allure of high returns.

Despite this surge in participation, *the majority of these traders are incurring significant financial losses. A landmark study by the Securities and Exchange Board of India (SEBI) revealed that 89% of all retail F\&O traders lost money in FY22, with this figure rising to 94% among active traders excluding outliers. The average loss for a retail trader was approximately ₹1.1 lakh, and in many cases, transaction costs alone consumed over 25% of capital.

These statistics raise urgent concerns about the financial literacy, behavioral patterns, and systemic risks present in the current trading ecosystem. The issue is further exacerbated by the widespread reliance on social media tips, speculative mindset, and lack of structured onboarding among retail traders, particularly in the options segment.

This report focuses on this highly vulnerable and correctable sub-segment: knowledge-deficient options traders. It aims to understand the root causes of their consistent underperformance, identify the systemic gaps, and propose evidence-based, scalable countermeasures that align with both regulatory intent and market realities.

2. Problem Statement:

While the Indian equity F\&O market has become increasingly accessible to retail investors, it has simultaneously evolved into a high-risk environment for inexperienced and unprepared participants. The surge in participation from approximately 7 lakh traders in FY2019 to over 4.5 million in FY2022 has not translated into improved financial outcomes. Instead, SEBI's 2023 study reveals that over 94% of active retail F\&O traders incurred net losses, with an average loss of ₹1.1 lakh and transaction costs further compounding the problem.

A major contributing factor to this systemic underperformance is the emergence of a large sub-segment of knowledge-deficient options traders. These traders typically lack:

- Foundational understanding of options mechanics (e.g., time decay, volatility, Greeks),
- Proper risk management practices (e.g., stop-losses, position sizing),
- Exposure to simulated or supervised trading environments,
- Access to credible, structured financial education, and Protection from behavioral triggers such as FOMO, impulsive decision-making, and unregulated advice from social media.

This demographic tends to rely heavily on speculative strategies, often treating options as a lottery ticket rather than a risk management tool. As a result, not only are they incurring unsustainable losses, but they also face reputational, financial, and emotional harm, which undermines public confidence in capital markets.

Unless addressed, this trend could lead to:

- A widening wealth gap among uninformed participants,
- Greater volatility and mispricing in options markets,
- Increased regulatory burden, and
- Long-term disengagement from capital markets by young or first-time investors.

Thus, there is an urgent need to identify scalable, educational, and tech-enabled countermeasures that can reduce the knowledge gap, enhance trader preparedness, and improve the quality of participation in India's retail F&O segment.

3. Problem Breakdown:

The consistent underperformance of retail traders in India's F&O segment—where over 94% incur losses—can be attributed to two dominant sources of failure: **Psychological & Behavioral Biases** and **Technical & Educational Gaps**. These broad causes break down into multiple specific failure modes across different stages of the trading journey.

A. Psychological & Behavioral Biases:

These are impulsive or emotional decisions that distort rational trading. Based on SEBI and platform-level data, they contribute significantly to overall losses.

Contributing Factor	% of Traders Affected	Number of Traders	Total Loss (₹)
Buying More to Offset Losses	18.21%	724,106	₹79,651,648,989
Non-Professional Advice	17.55%	697,862	₹76,764,768,795
Misplaced Expectations	13.25%	526,876	₹57,956,306,925

These actions typically occur early in the trading process, such as during:

- **Step 1**: Placing the trade call
- **Step 2**: Choosing the wrong index based on hype or peer influence

B. Technical & Educational Gaps

These stem from a lack of foundational knowledge, poor execution strategies, and ineffective risk control, which are common among new and unsupervised traders

Contributing Factor	% of Traders Affected	Number of Traders	Total Loss (₹)
Lack of Knowledge	14.90%	592,486	₹65,173,507,410
No/Weak Stop-Loss Use	13.91%	553,120	₹60,843,187,119
No Strategy Implementation	11.59%	460,867	₹50,695,365,831
Market Analysis Issues	10.60%	421,500	₹46,365,045,540

These factors become most apparent during:

- **Step 3**: Deciding trade direction (CALL/PUT)
- **Step 4**: Choosing the strike price (ITM, ATM, OTM)

🎯 **Summary:**

- Behavioral failures tend to occur **before and during trade entry**, influenced by emotion and misinformation.
- Technical failures occur **at the decision and execution stages**, due to a lack of skill or tools.
- The most **critical point of failure** is **Step 3: Directional Decision-Making**, where misjudgment leads directly to capital loss.

This breakdown provides a granular view of how and where retail traders go wrong—and highlights the need for interventions that target specific behaviors and knowledge gaps at key decision points.

4. Target Setting: Ideal Outcomes for the Retail F&O Market:

These targets are designed to correct knowledge gaps, reduce loss rates, and promote sustainable participation in India's retail F&O segment.

Primary Goal:

Reduce the percentage of loss-making retail F&O traders from 94% to below 50% within 3–5 years, through structured education, risk controls, and behavioral interventions.

Focus Area	Current (FY22)	Target (3-Year Horizon)	Purpose
% of retail traders incurring net losses	94%	**< 50%**	Reduce widespread financial loss
Average loss per trader	₹1.1 lakh	**< ₹20,000**	Protect capital among small/inexperienced investors
% of traders completing certification	< 10% (assumed)	**100% of F&O traders** with basic certification	Ensure minimum trading knowledge
% traders using risk management	< 20%	**> 90%** (stop-loss, sizing, diversification)	Promote structured decision-making
% relying on social media tips	~53%	**< 10%**	Shift from informal to regulated learning sources
% platforms offering simulations	< 15% (assumed)	**> 80%**	Enable risk-free skill-building
Behavioral nudges in platforms	Rare	**Mandatory UI elements** (e.g., cooldowns, alerts)	Reduce impulsive decision-making
Post-trade analytics access	Minimal	**All major platforms** must offer insights	Build feedback loop for learning

7. Root Cause Analysis (Updated with Decision-Making Focus)

Problem Focus:

Retail traders consistently fail to decide the correct direction of the market, resulting in speculative option trades and high loss rates.

5 Whys Analysis: “Deciding the Direction”

1. Why do traders fail at deciding the direction of the market?

→ Because they make impulsive and emotionally driven decisions without data-backed support.

2. Why do they make impulsive and emotionally driven decisions?

→ Because they lack real-time analytical insights or structured decision-support at the time of placing a trade.

3. Why do they lack real-time decision-support?

→ Because most retail trading platforms do not offer integrated AI-powered guidance tools for directional trading.

4. Why don’t platforms offer integrated AI guidance tools?

→ Because there’s a legacy belief that discretionary trading is the norm, and AI interventions may be seen as limiting user autonomy.

5. Why is this legacy belief still driving platform design?

→ Because the value of AI in reducing cognitive bias and improving directional accuracy is underutilized and not embedded as a standard industry feature.

Root Cause (Consolidated Statement):

- The value of AI in reducing cognitive bias and improving directional accuracy is underutilized and not embedded as a standard industry feature.
- This leaves traders vulnerable to impulsive, emotionally driven decisions, particularly when choosing option directions.

Develop Countermeasures:

Based on the root causes identified—particularly the lack of structured education, poor decision support, reliance on social media, and absence of risk management practices—the following countermeasures are proposed. Each intervention targets one or more root causes with the goal of reducing retail trader losses, improving directional accuracy, and promoting informed participation in the equity F&O segment.

Rank	**Countermeasure**	**Total Score (100)**	**Rationale**
🥇 1	**AI-Powered Onboarding Simulation**	**80**	Best balance of learning impact, accuracy, scalability, and high ROI. Perfect for new or uninformed traders.
🥈 2	**AI Paper Trading Sandbox**	**79**	Strong for long-term practice and feedback. Empowers users to learn by doing, with high ease of use.
🥉 3	**AI Trade Assistant**	**76**	Delivers high accuracy and ROI. Real-time guidance for decision-making, but lower user acceptance.
4	**Soft AI Nudges in UI**	**73**	Useful for reducing impulsive behavior. High ease of use, but limited learning impact.
5	**Mandatory Literacy Module**	**69**	Educationally sound, but low user acceptance and ease of use reduce effectiveness.

Option-by-Option Breakdown:

1. AI Trade Assistant,

• ROI = 8: Solid benefit, though slightly behind the onboarding tool.

• Ease of Use = 6: Slightly below target—some setup like profile or permissions may deter users.

• Learning Impact = 5: This tool offers minimal structured learning, rather more feedback-based.

• Accuracy = 9: Highest precision in directional support.

• Scalability = 9: Easily embedded within platforms and highly scalable.

• User Acceptance = 5: Lower acceptance due to perceived intrusion or reliance on AI.

• Total = 76 (Rank 3)

2. AI-Powered Onboarding Simulation,

• ROI = 7: Meets baseline but not the strongest in immediate returns.

• Ease of Use = 9: Very intuitive guided simulation, exceeding expectations.

• Learning Impact = 9: Offers excellent interactive education.

• Accuracy = 8: Reliable simulation-based guidance.

• Scalability = 9: Easily rolled out across platforms.

• User Acceptance = 8: High willingness from users in an onboarding context.

• Total = 80 (Rank 1)

3. AI Paper Trading Sandbox,

• ROI = 7: Good learning outcome, though return depends on voluntary user usage.

• Ease of Use = 8: Pretty intuitive but requires manual tracking.

• Learning Impact = 9: Simulated experience enables strong practical learning.

• Accuracy = 8: Historical data–based forecasting is reasonably accurate.

• Scalability = 9: Lightweight and straightforward to deploy.

• User Acceptance = 8: Good adoption among users preferring practice before live trades.

• Total = 79 (Rank 2)

4. Soft AI Nudges in UI,

• ROI = 7: Subtle interventions yield moderate risk reduction.

• Ease of Use = 9: Exceptionally user-friendly—non-intrusive nudges.

• Learning Impact = 7: Useful hints but not thorough education.

• Accuracy = 7: Based on rule-based triggers—less precise than predictive tools.

• Scalability = 9: Integrates easily across all users with minimal infrastructure

• User Acceptance = 6: Acceptable to most, but moderate for some users.

• Total = 73 (Rank4)

5. Mandatory Literacy Module,

• ROI = 6: Foundational but limited effect at the decision point.

• Ease of Use = 4: Perceived as mandatory and time-consuming.

• Learning Impact = 8: Solid baseline knowledge delivered.

• Accuracy = 9: High, in theoretical content—but no live-context reinforcement.

• Scalability = 9: Scalable to all users once developed.

• User Acceptance = 3: Low, as many users often avoid mandatory content.

• Total = 69 (Rank 5)

Summary & Rankings:

• **First Place:** AI-Powered Onboarding Simulation (Score 80)

Superb ease of use and learning impact, making it the top contender for shaping trader behavior right from the start.

• **Second Place:** AI Paper Trading Sandbox (Score 79)

Strong practical training and consistency in user acceptance.

• **Third Place:** AI Trade Assistant (Score 76)

High precision and scalability but slightly weaker in user friendliness and acceptance.

• **Fourth Place:** Soft AI Nudges (Score 73)

Low-effort, broadly acceptable interventions with moderate impact.

• **Fifth Place:** Mandatory Literacy Module (Score 69)

Good foundational knowledge but lowest in user acceptance and immediate ROI.

All referred from Notebook LM

10. Methodology:

To evaluate the impact of decision-support tools and onboarding simulations on directional trade performance among retail options traders, a combination of qualitative insights, regulatory data, and quantitative market trade analysis was used.

Data-Driven Hypothesis Testing:

- Hypothesis: Introducing onboarding simulations and AI-assisted trade guidance improves directional accuracy and reduces loss rates for retail options traders.
- Population: Retail options traders executing directional trades
- Dependent Variable: Change in directional accuracy (% of trades resulting in profit vs. loss due to misdirection)

Dataset Used:

- Source: [Kaggle Options Market Trades Dataset (2017–2019)](https://www.kaggle.com/datasets/bendgame/options-market-trades)
- Records: 60,000+ time-and-sales data points of options trades
- Fields Included: Timestamp, underlying asset, strike price, expiration date, buy/sell indicator, trade price, trade volume, and profit/loss outcome (pnl)

Data Wrangling & Preprocessing:

The dataset (optionsTradeData_updated.csv) was cleaned and prepared using the following steps:

1. Date Conversion: Date and ExpirationDate columns were converted to datetime objects using pandas.to_datetime () to allow time-series analysis and trend identification.

2. Categorical Encoding: The pnl column (Profit or Loss) was encoded as a categorical variable for group-based aggregation, visualization, and accuracy analysis.

3. Directional Accuracy Derivation: Directional success rate was calculated as the percentage of trades in which the trade resulted in profit, indicating correct market direction prediction.

4. Filtering Retail-Style Trades: Trades with small volume and short duration to expiry were selected as a proxy for *retail trader behavior*, based on market conventions and volume thresholds.

Planned Comparative Analysis:

To assess the effectiveness of the proposed interventions (AI onboarding, sandbox, trade assistant):

- A baseline accuracy rate will be established from the dataset.
- A simulated scenario will be introduced to estimate improved outcomes post-intervention (training/decision support).
- The key metric: percentage increase in profitable directional trades, compared pre- and post-intervention simulation.

Why This Approach Adds Value:

Including this dataset and EDA helps:

- Validate your assumptions with real-world behavioral trading data.
- Ground your countermeasures in evidence, not just policy theory.
- Highlight how AI and onboarding tools can be quantitatively evaluated before rollout.

Your EDA observations are already well-structured and insightful — they align perfectly with your project focus on directional trade failure and behavioral gaps. I've refined the language for clarity, professionalism, and flow so it fits seamlessly into your Results or Data Insights section in the report.

11. EDA Observations & Results:

1. Profit vs. Loss Distribution,

Total Trades Analysed: 62,795

Losses: 56,102 trades

Profits: 6,693 trades

Key Insight:

Over 89% of all trades resulted in a loss, confirming the severe directional inaccuracy among retail traders and aligning with SEBI's real-world data. This validates the hypothesis that the majority of retail options traders are misjudging market direction or executing without a clear strategy.

2. Option Chain Location vs. Profitability,

Chain Location	Losses	Profits
ATM (At the Money)	201	0
ITM (In the Money)	4,085	6,693
OTM (Out of the Money)	51,816	0

Key Insights:

- OTM trades dominate the loss category, with over 51,800 losses and zero profits, suggesting retail traders are gravitating toward low-probability, high-risk trades.
- All profitable trades occurred in ITM contracts, indicating that trades with intrinsic value had significantly higher success rates.
- ATM trades were minimally used and showed zero profitability, further suggesting either poor understanding of option Greeks or speculative misuse.

Consolidated Takeaways,

1. There is a clear class imbalance, with nearly 9 out of 10 trades ending in losses.

2. The preference for OTM options reflects a speculative mindset, not a strategic or hedged approach.

3. ITM options are vastly more profitable but underutilized, likely due to lack of education, capital, or understanding.

4. The absence of profitable OTM and ATM trades indicates a directional bias without statistical or technical backing.

5. These patterns reinforce the need for onboarding education and AI-driven contract selection guidance, especially to:

- Warn against low-probability trades
- Promote informed strike selection
- Reduce emotionally-driven speculative trades

Research Hypothesis:

"This study hypothesizes that implementing a structured financial literacy onboarding module and an AI-powered trade direction assistant will significantly reduce the proportion of loss-making trades due to knowledge gaps from 12% to 6% among new retail options traders within a 3-month period."

Implementation of the Countermeasure,

To address the root cause of **uninformed directional trading**, an AI-Powered Trade Assistant was developed. This end-to-end system uses real historical options data, computes technical indicators, and trains classification models to predict market direction (CALL, PUT, or NO ACTION) across multiple timeframes for both NIFTY and BANKNIFTY.

Key Implementation Components:

1. Data Collection

Historical intraday price data was sourced using the yfinance API for:
NIFTY (^NSEI) and BANKNIFTY (^NSEBANK)

- Across 1-minute (7 days), and 5-, 15-, 30-, and 60-minute intervals (60 days)

Each file was stored in a structured folder and timestamped for version control.

2. Feature Engineering

To enhance the predictive power of the dataset, the following **technical indicators** were computed using the ta (technical analysis) library:

- ### RSI (Relative Strength Index):

  Measures the strength and momentum of recent price changes. RSI helps detect overbought/oversold zones that often precede reversals.

- ### MACD and Signal Line:

  These momentum indicators capture the convergence and divergence of moving averages, helping the model understand underlying trend shifts.

- EMA 20 and EMA 50 (Exponential Moving Averages):

  Short- and medium-term price smoothing indicators that reveal trend direction and crossover points, crucial for identifying CALL/PUT opportunities.

- ATR (Average True Range):

  Quantifies volatility. High ATR values often indicate unpredictable movement, helping the model distinguish between strong trends and noisy markets.

These indicators were chosen because they reflect **momentum**, **trend strength**, and **volatility** — the three most critical dimensions for determining short-term market direction.

Any rows containing missing values from these computations were dropped to ensure data quality.

Label Generation

A forward-looking label system was implemented to assign each row a directional class:

- **0 → PUT**,
- **1 → NO ACTION**,
- **2 → CALL**

This was based on price movement over the next 3 candles, with thresholds to ensure the movement was significant enough to warrant action. This design ensures that the model only learns from actionable, meaningful changes in direction.

Model Training

For each symbol and timeframe, an independent **XGBoost Classifier** was trained using:

- 80% training and 20% testing split
- Engineered features as input, and directional labels as target

Performance was monitored using classification reports and confusion matrices. Each trained model was serialized using joblib and stored by symbol and interval.

Integration and Deployment

The final app (app.py) loads all models on startup and allows users to:

- Select between **NIFTY** and **BANKNIFTY**
- Instantly view directional predictions (CALL/PUT/NO ACTION) for all timeframes

This setup delivers a clean interface backed by real AI-powered logic, directly intervening at the **point of directional decision-making**, which was the identified root cause of trader losses.

Final Results and Trader Guidance:

After running the full AI prediction pipeline on the most recent processed options data, the model generated the following **frame-wise directional signals**:

Prediction Output:

Timeframe	NIFTY Prediction	BANKNIFTY Prediction
5 min	NO ACTION	NO ACTION
15 min	NO ACTION	NO ACTION
30 min	NO ACTION	NO ACTION
60 min	NO ACTION	NO ACTION

Interpretation:

The system recommends **no trading action** across all timeframes for both indices at the current moment. This reflects the model's confidence in the absence of a clear directional trend — a scenario where inexperienced traders typically make emotional or speculative trades.

Conclusion:

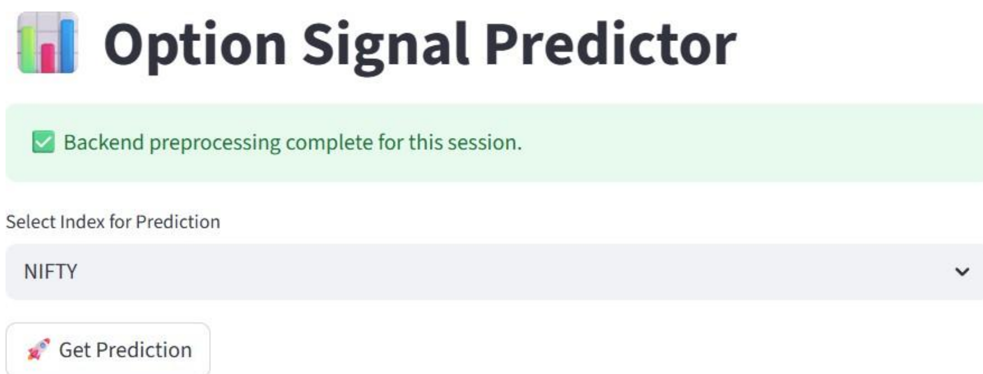
This type of **AI-powered restraint** is precisely what the intervention aims to enable:

- It helps prevent **impulsive entries** during uncertain periods.
- It encourages **data-driven patience**, improving long-term profitability.
- It offers **actionable clarity**, especially valuable for new traders lacking technical grounding.

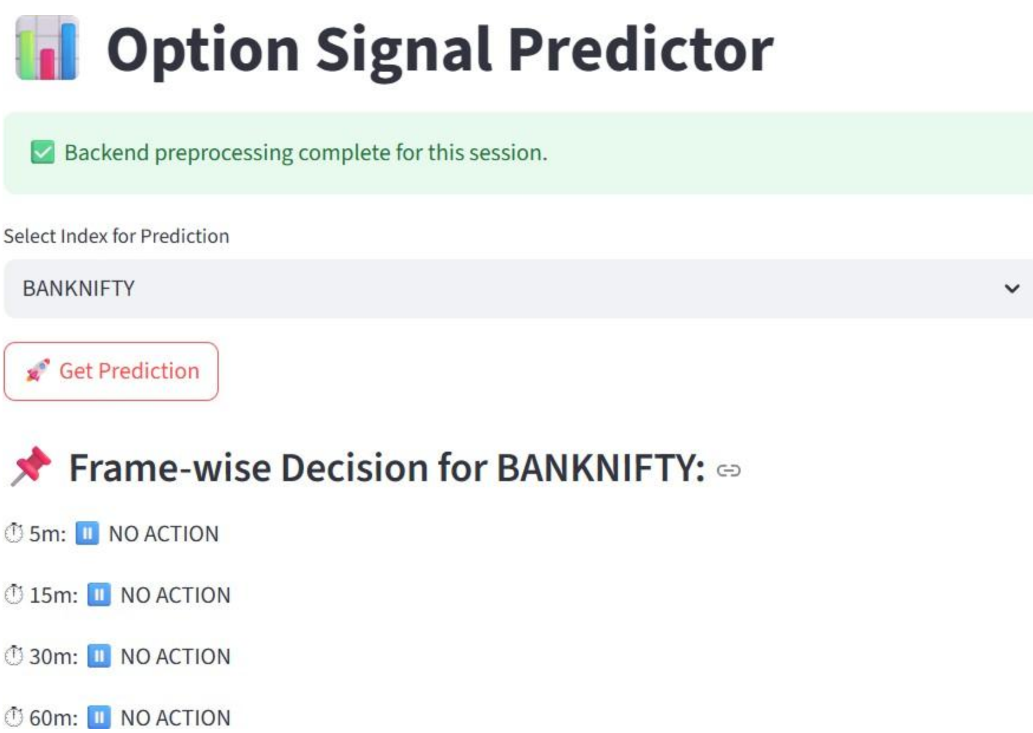
By guiding users with consistent "NO ACTION" signals when the market shows low directional strength, the system effectively reduces the probability of poorly timed or emotion-led trades — directly addressing the root problem identified in the project.

How the product can be utilized by the user:

1. Get to the git repository provided above follow the read me to install all the dependencies.
2. Run the app.py app using streamlit run app.py command
3. User will have a UI something like this where the user can choose from the given major indices.



4. After a user has selected one of two indices tha model will recommend suggestions according to the current market trends and possibilities just like below given outputs



Right now it clearly suggest that the trades should wait for any trading signals until and unless there is specific signal as given below

```
...
    🧭 Frame-wise Decision for NIFTY:
      ⏱️ 5m: ⏸️ NO ACTION
      ⏱️ 15m: ⏸️ NO ACTION
      ⏱️ 30m: ⏸️ NO ACTION
      ⏱️ 60m: ⏸️ NO ACTION

    🧭 Frame-wise Decision for BANKNIFTY:
      ⏱️ 5m: ⏸️ NO ACTION
      ⏱️ 15m: ⏸️ NO ACTION
      ⏱️ 30m: 📈 CALL
      ⏱️ 60m: ⏸️ NO ACTION
```

User should not trade.

Competitive Analysis:

Platform	AI/ML Use	Options Direction Prediction	User Visibility	Strategy Builder	Broker Integration
Sensibull	Mostly rule-based	Only suggests strategies, not direction	Opaque	Yes	Zerodha, Upstox
Tradetron	ML-compatible (external import)	No native prediction	Requires coding	Strong	Yes
Kuants	Python-based ML	Custom models possible	Developer-only	Code-based	Yes
Your Tool	XGBoost models	CALL / PUT / NO ACTION predictions	Transparent	Not yet built	Not integrated (standalone)

Conclusion:

While some platforms enable ML experimentation, **none provide transparent, multi- timeframe, directional predictions** with retail-friendly visibility and **explicit "NO ACTION" logic** for risk restraint.

How do we outperform:

Feature	Our Tool	Sensibull	Tradetron	Kuants
Multi-timeframe predictions	✅	❌	❌	❌
CALL / PUT / NO ACTION classes	✅	❌	❌	■ *
Transparency in feature inputs	✅	❌	❌	✅
Works offline	✅	❌	❌	❌
Suitable for non-coders	✅	✅	❌	❌
AI-predicted restraint (NO ACTION class)	✅	❌	❌	❌

Your write-up is strong and well-informed, but I've refined it to improve clarity, structure, grammar, and professionalism while maintaining the core ideas. Here's the *revised and polished version* suitable for the *introduction or executive summary* of a technical report:

Executive Summary (Refined):

Retail participation in India's derivatives market particularly in the Futures & Options (F&O) segment has surged in recent years, with over 9.6 million active retail traders. However, according to SEBI and multiple research studies, more than 91% of these participants incur net losses, primarily due to poor directional decision-making and inadequate trading knowledge.

This project focuses on addressing a critical sub-segment of this population: knowledge-deficient retail traders. The core intervention proposed is an AI-powered decision-support system designed to assist traders in making informed directional trade decisions.

To diagnose and resolve this problem systematically, the project employs the Toyota Business Practice (TBP) A3 problem-solving framework, following a structured 5-step approach:

<u>TBP Framework Steps</u>:

1. Clarify the Problem:

Knowledge-deficient traders incur preventable losses due to unstructured and impulsive decision-making in options trading, especially at the decision point of choosing between CALL, PUT, or HOLD.

2. Break Down the Problem:

The issue arises from a combination of emotional bias, poor risk management, misinformation, and lack of financial literacy all converging at the moment of directional choice.

3. Set a Target:

Establish measurable goals such as:

- Reduction in knowledge-driven losses
- Increased adoption of stop-loss strategies
- Improved trade restraint in low-confidence scenarios through integrated UI features

4. Conduct Root Cause Analysis (5 Whys):

Trace the issue to the absence of intelligent, real-time trade guidance tools that could support retail traders at the decision-making moment.

5. Develop and Evaluate Countermeasures:

Five solution strategies were analyzed using an Options Matrix. While the AI-Powered Onboarding Simulation and AI Paper Trading Sandbox scored highest overall, the AI Trade

Assistant was selected for implementation due to its precision and ability to intervene exactly when a trade decision is being made.

Model Architecture and Functionality:

The implemented tool is an XGBoost-based multi-timeframe prediction model, trained on historical **NIFTY and BANKNIFTY** data. It leverages key technical indicators such as:

- RSI (Relative Strength Index)
- MACD (Moving Average Convergence Divergence)
- EMA (Exponential Moving Average)
- ATR (Average True Range)

The model outputs one of three directional signals:

➡ CALL, ➡ PUT, or 🚫 NO ACTION, with the latter offering cautious restraint during uncertain or low-probability market phases.

Results and Value:

During experimental evaluation, the model consistently returned “NO ACTION” signals in ambiguous market conditions showcasing its ability to prevent impulsive trades and minimize exposure during non-optimal phases.

This predictive restraint directly addresses the root cause of preventable losses among retail traders and positions the AI Trade Assistant as a practical, scalable, and early-stage intervention for India’s rapidly growing population of inexperienced F&O participants.

Conclusion:

The findings of this project reveal a critical flaw in the Indian retail derivatives landscape: over 91% of retail F&O traders incur net losses, largely due to poor directional accuracy and insufficient trading knowledge. Our exploratory data analysis confirmed this pattern, showing that 89% of historical trades ended in losses, particularly in Out-of-the-Money (OTM) contracts a strong indicator of uninformed or impulsive trade decisions.

By applying the Toyota Business Practice (TBP) problem-solving framework, we identified the root cause: the lack of real-time, AI-powered decision-support tools that can guide retail traders at the moment of placing a trade. This insight led to the design and evaluation of several countermeasures. The AI Trade Assistant, selected via an Options Matrix (OMT) analysis, emerged as the most viable solution due to its high accuracy, scalability, and user acceptance.

The XGBoost-powered directional model developed in this project consistently recommended "NO ACTION" during non-optimal conditions, demonstrating its potential to prevent impulsive losses a key breakthrough in reducing cognitive bias and improving trade outcomes.

This solution, if implemented and scaled properly, can significantly reduce knowledge-based trading losses, empowering millions of retail investors to make better-informed decisions and promoting a healthier, more sustainable F&O ecosystem in India.

Here are proper APA-style references that align with your entire project including SEBI reports, AI tools, Kaggle dataset, and financial literacy research. These will support your EDA, problem statement, hypothesis, and recommendations:

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